The Exchange Lock and Its Key

Spin isomers of hydrogen and deuterium, what a conversion catalyst must supply, and when stored order is entanglement

On DIY Quantum Computing (working title) · laboratory note with Lean kernel K1

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The Lean kernel K1, its build and audit, the computations and this companion note were developed with Claude (Anthropic), working within MetaLabe.

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Contribution and responsibility. Lu Semita / EmergenceByDesign supplies the originating research program and its question: a low-qubit quantum computation that each adjudicator can build cheaply and see first-hand. The program also supplies the receipt-standard architecture, the conceptual framing and pedagogy, and continuing adjudication. Claude (Anthropic) contributed materially to this note: the formalization, the Lean kernel and its audit, the computations, the imported literature, and the wording. Claude accepts responsibility for those contributions as presented here. The attribution records contribution and responsibility; it does not assert separate legal personhood or ownership.

Abstract

This laboratory note seals in Lean 4 the relations that let molecular hydrogen and deuterium hold stored nuclear-spin order. It also states exactly what any agent must supply to turn one spin isomer into the other.

Exchanging two identical nuclei multiplies every state by a fixed sign, so the spin family fixes the rotation family. For hydrogen the antisymmetric spin family (para) rotates evenly and the symmetric family (ortho) oddly; for deuterium the assignment is mirrored. Three results follow for couplings.

- A coupling that is the same at both nuclei converts nothing.
- A coupling that tells the two spins apart but not the two positions has no effect inside a sector of definite statistics.
- Only a coupling that is odd under both exchanges converts, and the kernel proves that no other kind can: the field must differ across the bond.

Conversion also needs a sink that takes up at least twice the rotational constant in energy, together with the change in total spin. A uniform readout cannot recognize the isomer. A total-spin readout can, and it converts nothing. By Lagrange's identity, every product state and every finite mixture of product states has singlet weight at most one half, so a singlet weight above one half certifies entanglement. Two worked models, one for each statistics, show that none of these statements is empty.

The kernel holds 175 named declarations (114 theorems). It compiles with no errors and no warnings at trust level 0. The lock and the key use no axioms at all, and nothing uses classical choice. Three corpus modules are reused byte for byte, and the frame kernel and the seam kernel are shown to state the same proposition.

Lean certifies the model, and every physical identification is named as a bridge. One computed consequence shapes the build: the equilibrium para fraction crosses one half at 77.84 K. Liquid nitrogen at 77.36 K therefore sits at the edge, and a pumped nitrogen bath gives a margin of about ten points.

Conventions

Status labels. Labels record provenance and scope. They never record rank, and every work cited stands on equal footing and is checked against the others.

- Proved proved here from the stated definitions and hypotheses and checked by the Lean kernel.
- Reused a theorem from earlier Lean files of this corpus, imported unchanged and checked again in this build.
- Bridge a premise linking the Lean model to a physical system, argued here and not certified by Lean.
- Computed a finite numerical computation, kept with its script and output in the supplement.
- Imported outside correspondence or a measured number, with its source.
- Open not settled by anything here or found in the imports.
- Hypothesis a stated empirical hypothesis, with a test.

Kernel reports. After each Lean declaration the axioms reported by `#print axioms` are listed exactly. Three profiles occur besides the empty one. `propext` enters through rewriting with equivalences in `omega` and `simp`. `Quot.sound` enters through function extensionality, which the worked models use to compare states. `Classical.choice`, `sorry` and `native_decide` occur nowhere. The checker is Lean 4 [38]. The complete source, the build receipts and the one-paste bundle are in the supplement [7], hosted in the corpus repository and linked here rather than reproduced. Appendix A ties this note to those files by their SHA-256 hashes, and Appendix C gives every kernel report verbatim.

The record of its own making. Appendix D records how this note was produced, including the errors found and fixed. The PDF also carries its own source files as attachments: the LaTeX source, the figure script, the build script and the kernel report.

1. Origin, scope and method

Declared origin. Description begins at the first cut: a source role and a target role, a self-side and an other-side, that make a perspective in a space. The cut is declared, not posited as fundamental, and neither side comes first. The corpus anchors are `reference_polarity`, `reflexive_reference_preserves_role_distinction` and `no_first_position`, all from the frame kernel [2]. The last holds its premises as fields of a declared bridge structure. Section 4.2 applies the same cut to two identical nuclei.

Time. The kernel fixes no dimension count and does not name time "the fourth dimension". Time enters only as one representation within the construct, and from inside it the time direction closes only approximately.

Scope of "everything". The scope is a theory of everything that can be stated within the domains that use the same formalities of language and description. A question such as the Sphinx's "Are the gods real?" gets its answer from what its terms and their relation are taken to be, and that lies outside this scope.

Method at the end point. We begin in a middle we do not exclude, the first cut, and compare by parallax. That is, we adopt other axioms and check whether they get behind it. Inside the kernel, one collision theorem is read in four domains: role order, isomer recognition, turn parity and swing phase. It is used in its seam form (`universal_class_is_closed` [1]) and its frame form (`target_difference_blocks_factorization` [2]). Each reading names the added coordinate that would be adequate.

What Lean certifies. Lean certifies the model: the propositions in the file under their stated hypotheses. Whether a physical system satisfies those hypotheses is a bridge (Section 8). The model is checked against the prior corpus Lean in Section 2.

2. Established and reused

Module Reused	Used for	Original check
`UniversalRecoveryClosure`	the collision argument behind every "no reading recovers" result	Lean 4.22.0, trust level 0, every core axiom report empty [1]
`ClosureConsequences`	`Recovers`, `universal_class_is_closed`, `adequate_coordinate_must_distinguish`, `identity_readout_recovers`	as above [1]
`LambdaSpatiotemporalFrame`	reference polarity, the factorization obstruction (`FactorsThrough`)	Lean Web 4.33.0, 23 theorems, every axiom report empty [2]

Table 1: Corpus Lean reused unchanged.

All three modules are byte-identical to their sources (hashes in Appendix A) and were compiled again here. They reproduce their original counts: the R8 core gives 18 theorems and 25 named declarations, every axiom report empty, and the frame kernel's 23 theorems are again axiom-free.

How the kernel maps onto the prior corpus Lean.

- *Frame and seam.* `FirstCut.factorsThrough_iff_recovers` Proved shows that `FactorsThrough` of the frame kernel [2] and `Recovers` of the seam kernel [1] are the same proposition (`Iff.rfl`).
- *Collisions.* Every "cannot recover" result applies a reused theorem directly. Role order uses `target_difference_blocks_factorization`. Isomer recognition, turn parity and swing phase use `universal_class_is_closed`. Each "adequate coordinate" result uses `adequate_coordinate_must_distinguish`.
- *Edge-on reading.* `EdgeOn.fold_symm` and `edge_on_cannot_recover_phase` are the index form, on n evenly spaced phases, of three Part II theorems [3]: `inversion_preserves_spectral_read` ($\lambda(z^{-1}) = \lambda(z)$), `spectral_collision_has_two_branches` ($z = w$ or $zw = 1$) and `spectral_read_cannot_recover_phase`. The count of distinct swings, $\lfloor n/2 \rfloor + 1$, is Part II's

count. The step between the two forms is a bridge (M1, Section 8); Part II's theorems keep their own Mathlib receipts.

- *Closure.* `Closure.calendar_round` is a numeric instance of Part II's `simultaneous_return`, the joint return at the least common multiple [3]. `Closure.pythagorean_nonclosure` is a Lean check of the statement in *Increment, Registry, and Measure* [6, §17] that no finite number of exact fifths equals an integer number of octaves.
- *Sink, as correspondence only.* Part II §8 [3] records that a shared level splits at first order while an isolated level only shifts at second order. A real conversion therefore needs a partner state at the same total energy, which is the sink of Section 5.1. The low-field anomaly (H1, Section 10) is read against this correspondence as a hypothesis.

3. Established and imported

Table 2: Outside correspondence and measured numbers used in this note.

Item Imported	Number or statement	Used for
Room-temperature liquid sample carrying para-hydrogen order [11]	Para-H_2 prepared at 20 K over a charcoal catalyst ("essentially pure para"); sample at 295 K in a 400 MHz spectrometer. Purity 0.916 ± 0.019, entanglement of formation 0.822 ± 0.039. Matching it by direct cooling would need a 6.4 mK spin temperature or a 0.45 MT field.	The target computation has been run (low qubit count, entangled, sample at room temperature)
Algorithms on those states [12,13]	Deutsch's algorithm; Grover search over four items, with singlet purity 1.06 ± 0.04	same
Garage-scale enrichment [14]	Hydrated iron(III) oxide in a copper spiral in a liquid-nitrogen bath: 48.1–48.4 % para at 150 sccm, about 48–49 % up to 1 L/min	Cheapest enrichment route; it lands just below the one-half line
Conversion needs an inhomogeneous field [15,16]	"a homogeneous magnetic field should not affect the conversion"; the dipolar rate scales as $\mu^2 d^{-6}$	Matches Theorems 5 and 6
H_2 levels [17,18,19]	$J{=}0 \to 1$ interval 118.48677 cm^{-1} [17]. Raman lines as used in the supplement's script: $S(0) = 354.37$ and $S(1) = 587.03$ cm^{-1}, quoted as 354.36 and 587 in [18]; $S(2) = 814.42$ and $S(3) = 1034.67$ cm^{-1} for $J = 4, 5$, whose Boltzmann weight below 100 K is under 10^{-9}, so no reported digit depends on them. $B_0 = 59.322$, $D_0 = 0.0471$ cm^{-1}; D_2 $B_0 = 29.904$ cm^{-1} [19]	$p(T)$, $q(T)$ and the rotational gap
Separability edge [25,26,27,28]	Werner: a two-spin Werner state with singlet weight $\leq \frac{1}{2}$ is separable. Hildebrand: absolutely-PPT conditions for 3×3. Ishizaka–Hiroshima, Verstraete–Audenaert–De Moor: largest entanglement reachable by global unitaries	The other edge of the one-half line; the D_2 boundary
Spin–statistics [23]	The exchange sign equals the sign after one full turn	Held as a bridge structure in Lean, not proved
Two-turn sign seen by interference [29,30]	Neutron interferometry shows the sign change of a spinor under one full turn	Correspondence for Theorem 11

Item Imported	Number or statement	Used for
Room-temperature two-qubit gate in diamond [31,32]	Conditional gate between an NV electron spin and its nucleus; ^{14}N hyperfine $A_{\parallel} = -2.16$ MHz	C4, Section 9

4. Constructed here: the lock

4.1 Setting

A *signed carrier* is a set V with a map $v \mapsto -v$ and an element 0 such that $-(-v) = v$, and $-v = v$ only for $v = 0$. The two signs $\{+, -\}$ multiply as usual and act on V by $(+) \cdot v = v$ and $(-) \cdot v = -v$; they are the two square roots of one. An *exchange pair* adds two maps: the spin exchange P, an involution with $P(-v) = -P(v)$, and the position exchange Q, the half turn of the molecule. For signs $\sigma, \varepsilon, \eta$:

$$\mathrm{Stat}_\sigma(v) : \ PQv = \sigma v, \qquad \mathrm{Spin}_\varepsilon(v) : \ Pv = \varepsilon v, \qquad \mathrm{Rot}_\eta(v) : \ Qv = \eta v.$$

Stat_σ is the statistics of the pair (minus for two protons, plus for two deuterons). $\mathrm{Spin}_\varepsilon$ is the spin class (plus symmetric, minus antisymmetric). Rot_η is the rotation parity (plus for even J, minus for odd J). This is the standard account of the ortho and para forms [20–22], restated so that Lean can check it.

Theorem 1 (The isomer lock) Proved**.** *If* $\mathrm{Stat}_\sigma(v)$ *and* $\mathrm{Spin}_\varepsilon(v)$*, then* $\mathrm{Rot}_{\sigma\varepsilon}(v)$*.*

Proof. $Qv = P(PQv) = P(\sigma v) = \sigma P v = \sigma\varepsilon v$. □

Lean: `DIYQC.ExchangePair.isomer_lock`. Kernel report: no axioms

Corollary 1.1 (The four cases) Proved**.** *For a fermion pair, the antisymmetric family rotates evenly and the symmetric family oddly. For a boson pair, the symmetric family rotates evenly and the antisymmetric family oddly.*

Lean: `DIYQC.ExchangePair.fermion_antisymmetric_even`, `DIYQC.ExchangePair.fermion_symmetric_odd`, `DIYQC.ExchangePair.boson_symmetric_even`, `DIYQC.ExchangePair.boson_antisymmetric_odd`. Kernel report: no axioms (each)

Pair	Statistics σ	Spin family	Rotation	Name and spin weight
H_2 (two spin-$\frac{1}{2}$)	$-$	antisymmetric ($I = 0$)	even J	para, 1 state
	$-$	symmetric ($I = 1$)	odd J	ortho, 3 states
D_2 (two spin-1)	$+$	symmetric ($I = 0, 2$)	even J	ortho, 6 states
	$+$	antisymmetric ($I = 1$)	odd J	para, 3 states

Table 3: The lock. The rotation column follows from Theorem 1. The weights are Theorem 9.

4.2 The first cut and identical carriers

A pair (a, b) has two roles. The exchange $\mathrm{swap}(a, b) = (b, a)$ is an involution. A reading r is *role-blind* when $r(\mathrm{swap}\, p) = r(p)$ for all p.

Theorem 2 (The cut is held, not read) Proved**.** *For any carrier* x*, the source role and the target role on* x *are distinct. For any role-blind reading* r *and any two different carriers* $a \neq b$*, no function of* r *recovers which carrier holds the first role. A labeled reading (the identity) does recover it.*

Lean: `DIYQC.FirstCut.cut_is_held_not_read`, `DIYQC.FirstCut.role_blind_reading_cannot_order`, `DIYQC.FirstCut.labelled_reading_recovers_order`. Kernel report: no axioms

In physics terms, the exchange constraint exists because the two nuclei are identical. With HD the carriers can be told apart, the order is readable, and nothing locks. The cut is made by the describer, and the described pair does not carry which side came first.

5. Constructed here: the key

A *coupling* is a map $A : V \to V$ that commutes with the sign. It has *spin parity* s when $APv = s\,PAv$ for all v, and *position parity* t when $AQv = t\,QAv$ for all v.

Theorem 3 (How a coupling moves the classes) Proved. *If A has spin parity s and position parity t, then $\mathrm{Spin}_\varepsilon(v) \Rightarrow \mathrm{Spin}_{s\varepsilon}(Av)$ and $\mathrm{Stat}_\sigma(v) \Rightarrow \mathrm{Stat}_{ts\sigma}(Av)$. Two different spin classes, or two different statistics, meet only at 0.*

Lean: `DIYQC.ExchangePair.spinClass_of_parity`, `DIYQC.ExchangePair.statistics_of_parity`, `DIYQC.ExchangePair.spinClass_disjoint`, `DIYQC.ExchangePair.statistics_disjoint`. Kernel report: no axioms

Theorem 4 (The three cases) Proved. *Let $\mathrm{Stat}_\sigma(v)$ and $\mathrm{Spin}_\varepsilon(v)$.*

(i) If A commutes with spin exchange, Av stays in class ε; whatever it sends into the other class is 0.

(ii) If A anticommutes with spin exchange but commutes with position exchange, Av has statistics $-\sigma$; whatever it leaves in statistics σ is 0.

(iii) If A anticommutes with both, Av keeps statistics σ, lies in class $-\varepsilon$, and has rotation parity $\sigma(-\varepsilon)$: the rotation parity flips.

Lean: `DIYQC.ExchangePair.even_cannot_convert`, `DIYQC.ExchangePair.spin_odd_pos_even_leaves_sector`, `DIYQC.ExchangePair.spin_odd_pos_even_vanishes_in_sector`, `DIYQC.ExchangePair.key_converts`. Kernel report: no axioms (each)

Theorem 5 (Only the key converts) Proved. *Let A have spin parity s and position parity t. Suppose v has statistics σ and class ε, and $Av \neq 0$ has statistics σ and class $-\varepsilon$. Then $s = t = -$.*

Lean: `DIYQC.ExchangePair.converting_coupling_is_doubly_odd`. Kernel report: no axioms

	commutes with position exchange	anticommutes with position exchange
commutes with spin exchange	converts nothing (4.i)	converts nothing (4.i)
anticommutes with spin exchange	no effect inside the sector (4.ii)	**converts: the key** (4.iii, 5)

Table 4: The four parity classes of a coupling. Only the doubly odd class converts.

Two-site couplings. Let A_1, A_2 be the same coupling applied at nucleus 1 and at nucleus 2, so that $A_1P = PA_2$ and $A_2P = PA_1$. Their sum is the part that is the same at both nuclei; their difference is the part that differs between them.

Theorem 6 (Only the difference converts) Proved. *$A_1 + A_2$ commutes with spin exchange and so converts nothing. $A_1 - A_2$ anticommutes with it and always moves a state to the other spin class.*

Lean: `DIYQC.SitePair.sum_even`, `DIYQC.SitePair.difference_odd`, `DIYQC.SitePair.sum_cannot_convert`, `DIYQC.SitePair.difference_flips`. Kernel report: no axioms

Read together, Theorems 4–6 say what a catalyst must supply. The coupling has to differ between the two nuclei, and inside a sector of definite statistics it has to differ between the two positions as well: the field must change across the 0.74 Å bond. This is the inhomogeneity condition of the paramagnetic mechanism [15,16], stated as a symmetry requirement on any agent whatever. The kernel fixes the symmetry the agent must supply, not which agent supplies it (Section 10).

5.1 The sink

Theorem 7 (Parity change costs at least $2B$) Proved. *With levels $E_J = B\,J(J+1)$: if $J < K$ then $E_K \geq E_J + 2B(J+1)$. If J and K differ in parity, the rotational energy changes by at least $2B$, and the minimum is attained between $J = 0$ and $J = 1$.*

Lean: `DIYQC.Sink.rotLevel_gap`, `DIYQC.Sink.parity_change_needs_energy`, `DIYQC.Sink.lowest_gap`. Kernel report: `propext`, `Quot.sound`

With the measured interval [17] the $J{=}0 \to 1$ gap is 170.48 K for H_2. The rigid-rotor $2B$ for D_2 is 86.05 K. By Theorem 4(iii) the key flips the rotation parity, so every conversion step must hand at least this energy to, or take it from, something else. It must also take up the change of total nuclear spin, from 0 to 1 for hydrogen (Theorem 8). The same coupling links the two isomers in both directions (Section 6), so the sink's temperature, not the key, sets the direction (bridge B5).

5.2 Recognition is not conversion

For two spin-$\frac{1}{2}$ nuclei, take the triplet states T_+, T_0, T_- and the singlet S. The *uniform reading* is twice the total z-spin, and the *total-spin reading* is 0 on S and 2 on the triplet.

Theorem 8 (Recognizing is a different act) Proved. *No function of the uniform reading recovers the isomer. Any added coordinate that, together with the uniform reading, recovers it must separate T_0 from S. The total-spin reading recovers the isomer. Conversion changes the total spin.*

Lean: `DIYQC.Recognition.uniform_reading_cannot_recognize`, `DIYQC.Recognition.adequate_coordinate_separates`, `DIYQC.Recognition.total_spin_recognizes`, `DIYQC.Recognition.conversion_changes_total_spin`. Kernel report: `propext`

A total-spin readout is exchange-even. By Theorem 4(i) it can recognize without converting.

5.3 Counting the families

Theorem 9 (Spin weights and the cold family) Proved. *For d levels per nucleus,* $\mathrm{sym}(d) + \mathrm{anti}(d) = d^2$ *and* $\mathrm{sym}(d) = \mathrm{anti}(d) + d$. *For H_2 ($d = 2$) the weights are $3 : 1$, and for D_2 ($d = 3$) they are $6 : 3$. Call the family that rotates evenly the cold family. For $d \geq 2$ it is a single state exactly for a pair of spin-$\frac{1}{2}$ fermions; for a boson pair it holds at least three states.*

Lean: `DIYQC.Counting.sym_add_anti`, `DIYQC.Counting.hydrogen_families`, `DIYQC.Counting.deuterium_families`, `DIYQC.Counting.cold_family_single_iff`. Kernel report: `propext`, `Quot.sound`

5.4 When stored order is entanglement

Write the amplitudes of nucleus 1 and nucleus 2 as Gaussian-integer pairs $a = (a_1, a_2)$ and $b = (b_1, b_2) \in \mathbb{Z}[i]^2$, and set $\|a\|^2 = |a_1|^2 + |a_2|^2$. The singlet amplitude of $a \otimes b$ is $(a_1 b_2 - a_2 b_1)/\sqrt{2}$.

Theorem 10 (The witness) Proved.

$$\|a\|^2\|b\|^2 = |a_1b_2 - a_2b_1|^2 + |a_1\bar{b}_1 + a_2\bar{b}_2|^2.$$

Hence every product state has singlet weight $|a_1b_2 - a_2b_1|^2/(2\|a\|^2\|b\|^2) \leq \frac{1}{2}$*, and so does every finite mixture of product states with non-negative weights. A singlet weight above one half excludes every such mixture.*

Lean: `DIYQC.Witness.lagrange`, `DIYQC.Witness.product_singlet_bound`, `DIYQC.Witness.separable_singlet_bound`, `DIYQC.Witness.above_half_is_entangled`. Kernel report: `propext`, `Quot.sound`

Extending the inequality from Gaussian-integer to complex amplitudes is bridge B6, and the finite-mixture form of separability is bridge B7. The converse, that a singlet weight at most one half leaves the Werner state separable, is imported [25].

6. Worked models for both statistics

Two finite models keep the general theorems from being empty. Each carries the spin space, the half turn (even-J amplitudes keep their sign, odd-J amplitudes change it), a uniform field (the z-coupling at each nucleus) and a field that differs across the molecule. The second has the same z-coupling, multiplied by a rotational factor that exchanges the even-J and odd-J manifolds.

Two protons (`Model.Hydrogen`)	Two deuterons (`Model.Deuterium`)
start: para, antisymmetric spin, even J, statistics −	start: ortho (total spin 0), symmetric spin, even J, statistics +
uniform field (sum): annihilates it	uniform field (sum): annihilates it
spin difference, no position difference: symmetric spin with even J, statistics + (wrong for protons)	spin difference, no position difference: antisymmetric spin with even J, statistics − (wrong for deuterons)
field difference across the molecule: ortho, symmetric spin, odd J, statistics −	field difference across the molecule: para, antisymmetric spin, odd J, statistics +
the same coupling leads back to para	the same coupling leads back to ortho

Table 5: What each kind of coupling makes from the cold isomer. Every entry is a Lean theorem (`Model.*`, kernel report `propext`, `Quot.sound`).

The models are rotational truncations, with even and odd J collapsed to one manifold each and only z-components kept. They witness the theorems; they do not simulate a molecule.

7. Symbols and their formal anchors

Pedagogy can cure the forgetting that a representation brings, and a symbol can also become the next pharmakon. Each symbol below is paired with what it carries, the theorem that anchors it, and what it does not carry.

Table 6: Pedagogical symbols, what each carries, its formal anchor, and what it does not carry.

Symbol	Carries	Anchor	Does not carry
Prince Rupert's drop	stored order held locked; a one-way release once the right seam is touched	Theorem 4(i) (no key, no conversion); Theorem 14 (a cascade that does not die)	the fracture mechanics; the catalyst key opens both ways, while the drop releases one way
Paired pool dimples	two surface readings are the two ends of one connected structure in a further degree of freedom	Theorem 12; Theorem 2	any claim that the surface is "really" two-dimensional, or that the further freedom is a hidden object
The lock to an antiparticle	paired opposite signs that the reading cannot separate	the sign algebra; Theorem 11	particle physics; the kernel's signs are exchange and turn parity only
Bose–Einstein [33,34]	many in one symmetric state	Theorem 9: the picture fails for the nuclear spins of D_2, whose cold family is six-fold	condensation thermodynamics
Belt or plate (two turns)	a sign invisible to a direction reading and visible against a reference	Theorem 11	the spin–statistics theorem itself (a bridge)
Calendars and the comma	closure in whole-number ratios; non-closure leaves residue	Theorem 13	any cosmology
A fire that stays lit	below threshold the error cascade dies out; above it, it does not	Theorem 14	the constants of any particular code

Theorem 11 (The two-turn return) Proved**.** *The sign of a spin-$\frac{1}{2}$ state after n full turns returns after two turns. A direction reading cannot recover it. Any adequate added coordinate, for example interference against a reference that did not turn, must separate zero turns from one turn. Under the spin–statistics bridge, a proton pair has exchange sign $-$ and a deuteron pair $+$.*

Lean: `DIYQC.TwoTurn.returns_after_two_turns`, `DIYQC.TwoTurn.direction_cannot_recover_sign`, `DIYQC.TwoTurn.reference_must_separate`, `DIYQC.TwoTurn.proton_pair_is_fermion`, `DIYQC.TwoTurn.deuteron_pair_is_boson`. Kernel report: `propext`, `Quot.sound` (the first); the others report no axioms

Theorem 12 (The edge-on swing) Proved**.** *For n evenly spaced phases, the swing position* $\mathrm{fold}(n,k) = \min(k \bmod n,\ n - k \bmod n)$ *satisfies* $\mathrm{fold}(n,k) = \mathrm{fold}(n, n-k)$ *for* $0 < k < n$*. When $k \neq n-k$, no function of the swing recovers the phase. For $n = 12$ the swing runs* $0,1,2,3,4,5,6,5,4,3,2,1$*.*

Lean: `DIYQC.EdgeOn.fold_symm`, `DIYQC.EdgeOn.edge_on_cannot_recover_phase`, `DIYQC.EdgeOn.swing_positions_12`. Kernel report: `propext`, `Quot.sound`

Theorem 13 (Closure in whole-number ratios) Proved**.**

$$\mathrm{lcm}(13,20) = 260, \qquad \mathrm{lcm}(260,365) = 18980 = 52 \cdot 365 = 73 \cdot 260, \qquad \gcd(260,365) = 5.$$

For $m > 0$, $3^m \neq 2^{m+n}$: no positive number of exact fifths closes on octaves. Twelve fifths against seven octaves leave $3^{12} : 2^{19} = 531441 : 524288$*. Finally, $w^2 + 3w^2 = (2w)^2$, the $2:1$ closure behind a selective gate (C4).*

Lean: DIYQC.Closure.calendar_round, DIYQC.Closure.pythagorean_nonclosure, DIYQC.Closure.pythagorean_comma, DIYQC.Closure.selective_closure. Kernel report: propext, Quot.sound

Theorem 14 (The threshold) Proved. *Let one round of layered correction map a scaled error count x to $\lfloor c\,x^2/N \rfloor$. If $c\,x < N$ each round strictly lowers the count, and it reaches 0 within x rounds. If $N \le c\,x$, no round lowers it.*

Lean: DIYQC.Threshold.step_lt, DIYQC.Threshold.below_threshold_dies_out, DIYQC.Threshold.above_threshold_no_decrease. Kernel report: propext, Quot.sound

This is the recursion behind the fault-tolerance threshold [35,36], read as a fire that stays lit only if it is fed above a line.

8. Bridges

B1 Bridge Two identical nuclei with spin and rotation form an exchange pair. Total exchange is spin exchange composed with position exchange, and the statistics sign comes from spin–statistics [23], held in Lean as `SpinStatisticsBridge`.

B2 Bridge Couplings act linearly. By standard linear algebra over the two commuting exchanges, every coupling splits into the four parity classes of Table 4.

B3 Bridge A catalyst's field at the two nuclei forms a site pair. "Differs across the bond" means that the difference part has position parity minus.

B4 Bridge Rotational levels follow $E_J = B\,J(J+1)$ in the theorems. The measured levels move the threshold temperature by about 0.1 K (C1).

B5 Bridge The equilibrium isomer fractions follow from Boltzmann weights with nuclear degeneracies 1 : 3 (H_2) and 6 : 3 (D_2). The direction of conversion follows detailed balance with the sink.

B6 Bridge Gaussian-integer amplitudes stand for complex amplitudes. The witness inequality is homogeneous and polynomial, so scaling and density carry it to all amplitudes.

B7 Bridge In finite dimension a separable state is a finite mixture of product states (Carathéodory [37]).

M1 Bridge The edge-on results are the index form of Part II's spectral-read results; $\cos(2\pi k/n)$ is injective for $0 \le k \le n/2$.

9. Computed

C1 Computed With the measured levels, the equilibrium para fraction of H_2 passes one half below 77.84 K. The rigid rotor gives 77.94 K and the distorted rotor 77.82 K. At 77.36 K the equilibrium margin is +0.33 points. A 77 K generator as measured (48 %) sits 2.0 points below the line. Pumping the bath to 65 K gives +10.53 points, and to the triple point 63.15 K gives +12.34 points (Figure 1a).

C2 Computed For D_2, global unitaries can make the spin pair entangled exactly when the ortho fraction exceeds 6/7. The ortho fraction passes 6/7 below 39.29 K. The best negativity is $(7q - 6)/12$, and the para-D_2 fraction never exceeds 1/3 (Figure 1b). This uses the imported Hildebrand conditions [26].

C3 Computed For a two-proton register: with the singlet order kept, it can be entangled for $p > 0.5$; with the order dephased, for $p > 0.78$, that is below 49.3 K [27,28].

C4 Computed For an NV electron-nuclear register with $A = 2.16$ MHz [32], a frequency-selective CNOT closes at $\Omega = A/\sqrt{3}$. The rectangular pulse beat the Gaussian pulse in 54 of 54 matched cases. The realistic-initialization Bell fidelity stays above 0.5 for line widths up to 2 MHz (Gaussian lines).

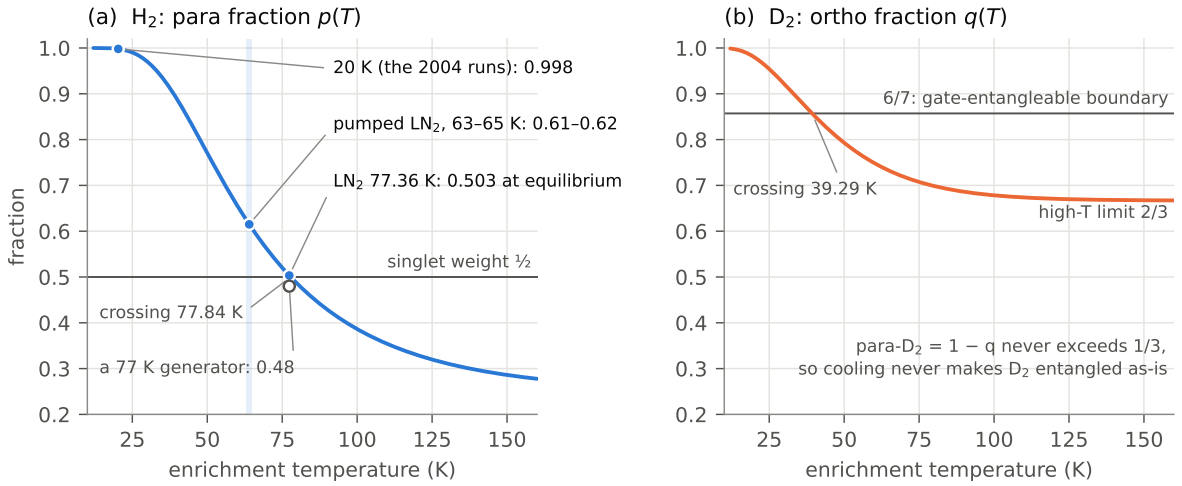


Figure 1: Equilibrium isomer fractions against enrichment temperature. (a) H_2 para fraction from measured levels; the line at one half is the witness threshold of Theorem 10, crossed at 77.84 K. Filled dots are equilibrium values at 20.3, 64 and 77.36 K; the open dot is the measured output of a 77 K generator [14]; the shaded band is a pumped nitrogen bath. (b) D_2 ortho fraction (rigid rotor); the line at 6/7 is the gate-entangleable boundary (C2). Scripts: `computations/para_fraction_threshold.py`, `paper/make_figures.py`.

10. Open, and one hypothesis

O1 Open Conversion rate magnitudes. Only the selection rule is proved.

O2 Open Over rare-earth catalysts, the conversion rate falls in fields below about 40 Oe. A 2025 review records that no explanation has been proposed [15].

O3 Open Conversion on ferromagnetic surfaces, whose fields are homogeneous at large scale, and on non-magnetic surfaces [15]. The kernel fixes the symmetry any agent must supply, not which agent supplies it.

O4 Open Rate changes under homogeneous external fields [15]. These can act through the catalyst's own state without contradicting Theorem 4. That reading is untested here.

O5 Open A Lean witness for the D_2 6/7 boundary, and the Werner converse.

O6 Open The cheapest readout, transfer chemistry and electronics for a garage register (rung 3, Section 11).

H1 Hypothesis The low-field drop recorded in O2 sits where Zeeman and rotational levels become shared (level crossings), where Part II §8 predicts first-order rather than second-order behavior [3]. Test: map the rate against field through the crossing fields.

11. The first-hand ladder

Each rung lets an adjudicator see one relation directly instead of taking the representation on faith. The engineering record will cost and source each rung.

0. *No cryogenics.* Bench analogues of the formal anchors: a belt or plate for the two-turn return, a rotating marker seen edge-on for the swing collision, gear trains for calendar closure, a monochord for the comma, and paired pool dimples. They demonstrate the anchors, not the isomers.
1. *Liquid nitrogen.* A copper coil packed with hydrated iron(III) oxide in a nitrogen bath gives about 48 % para [14]. This shows the key at work: without the catalyst nothing converts, and warmed para persists. The readout is a thermal-conductivity cell (an imported method, to be specified).
2. *Pumped liquid nitrogen.* Pumping the bath toward 63–65 K gives $p \approx 0.61$–0.62, about ten points above the line (C1). This is the cheapest rung at which the stored order is entanglement as-is (Theorem 10).
3. *Addressable qubits.* Transfer the singlet order to two inequivalent nuclei by para-hydrogen chemistry, then read and drive them by NMR. This was done in 2004 with para-hydrogen made at 20 K [11–13]. The cheapest garage version is open (O6).

   Safety. Hydrogen, cryogens and vacuum together, including oxygen condensing in cold lines that leak air, need a written protocol in the record before anyone builds.

12. Both edges

What is not claimed. The kernel does not prove that any physical sample satisfies B1–B7. It gives no conversion rates. It asserts no fundamentality, no first position and no hierarchy. "Room-temperature quantum computation" means a sample at room temperature carrying spin order made in a cold step: the cooling is moved in time, not removed.

What an objection has to show. Relabeling the objects is not a counterexample. An objection must name a failed hypothesis, an invalid bridge, or a step that is not preserved; this is structural transport in the Double-Edged Receipt Standard [4, Prop. 2]. Doubt about a claim is not a negative receipt either: until a receipt exists, the status is open, not false [4, Cor. 1]. The standard cuts both ways. It cuts a generator who over-claims, and it cuts a nullifier who turns finite non-observation into exclusion.

13. Lineage and acknowledgments

The state spaces in the kernel sit on the standard Hopf ladder $S^1 \to S^{2n+1} \to \mathbb{C}\mathrm{P}^n$ [24]. The unit vectors of $\mathbb{C}^{n+1}$ form S^{2n+1}, and the physical states (rays) form $\mathbb{C}\mathrm{P}^n$.

- One spin-$\frac{1}{2}$ is the first rung, $S^1 \to S^3 \to S^2$, where $S^2 = \mathbb{C}\mathrm{P}^1$ is the Bloch sphere.
- The two-proton spin space is $\mathbb{C}^4$, with rays $\mathbb{C}\mathrm{P}^3$ ($S^1 \to S^7 \to \mathbb{C}\mathrm{P}^3$).
- The two-deuteron spin space is $\mathbb{C}^9$ ($S^1 \to S^{17} \to \mathbb{C}\mathrm{P}^8$).

Theorem 11 reads on the first rung. A direction reading is the projection to the base S^2, and the turn sign lives in the S^1 fiber, which that projection forgets.

The same ladder carries the Hopf–Clifford geometry of Jenny Lorraine Nielsen's Topological Unified Field Theory [9] and of the joint registry study [10]. Her Hopf-fibration work is part of the incremental lineage of this corpus, including the Nielsen–Semita Attractor Framework, and

this note acknowledges it as such. No physical identification made in that theory is used as a premise here, and none is asserted.

The corpus works this note builds on are cited as accepted in their reported environments, with scope as declared [1–8].

A. The seal

Toolchain	Lean (version 4.35.0, x86_64-pc-linux-gnu, commit eb6cafb7290e8bd5f2937522d8d804f58f02bafc, Release). Compiled here from the official source tree at tag v4.35.0, because the release archive could not be downloaded from this environment. The commit is identical to the toolchain of the Axiom as Pharmakon R3 receipt [5].
Executable	`bin/lean` SHA-256 `8856bb7bf38e3d7a7bad2abebd60b630065027b922523605224743d82b76818f`, a launcher linked to `libleanshared.so`, SHA-256 `725d12c4780f07f75d0908ce35e16dfdd483d706a68f7bac757a62ae0d996091`
Library	bundled Lean libraries only (no Mathlib); trust level 0 for every compilation
Kernel	`DIYQCKernel.lean`, SHA-256 `ff1a56299b4f4b23d3962a359ba54ba3c132eb00be3cfa3c81c542228d83bbad`. 175 named declarations (114 theorems); no `sorry`, `admit`, `native_decide` or axiom declarations; 0 errors, 0 warnings
Axiom profiles	no axioms: 111; [`propext`]: 10; [`Quot.sound`]: 1; [`propext`, `Quot.sound`]: 53. All 59 declarations of the lock and the key (Sections 0–5 of the source) report no axioms
Reused	`UniversalRecoveryClosure.lean` `afe5ba209e90b4931f121875e7635b67bd5a0fcc9dc1b034fe622c93a2b24492`; `ClosureConsequences.lean` `f23628d530069d07e61244c2b85a69d28d29769c45e36107a2bcb8a1db91a304`; `LambdaSpatiotemporalFrame.lean` `f1769ecdf131297fef4fb1e2389166f4e78638753ae301b0dfc82c38c5be9904`; all unchanged
Sources	corpus repository [8], commit `55436de8367c06d81fc5d9a35a5ada503c1ce09a`: the R8 Complete Proofs archive (SHA-256 `dec8a3bb…f7eab89c`) and the frame-kernel PDF (SHA-256 `f73ad48b…4d01d868a`)
Hosted copies	folder `WIP_docs/IRRProofEngine/DERIVATIVE/DIY_Quantum_Computing/` of [8]: `DIYQC_Supplement_K1_2026-10-31.zip` `97e0063a2edcd96216222853723c91fd769dc561ced05e90130adfa6ad8744e0`; `DIYQC_Kernel_K1_bundle.lean` `754351c5c6cc8af7d8620afcabfd5c1ebab678af6a5d89acbba0e30f05dd4d48`; `DIYQC_Claim_Ledger_K1-1.md` `7072d74e09e532e9f9585a71d31390864c55f3e1a8b8c26521a1a69af503754e`. Each is byte-identical to the file checked here, and the kernel inside the archive has the hash given above
Rebuild	`python3 reproduction/verify_diyqc.py --lean /path/to/lean` in the unpacked supplement. The audit compiles each target in an isolated path, as the R8 script does [1]

B. Claim register

Machine-readable form: `THEOREM_REGISTER.json` in the supplement. Return rule: Carry each proved result forward in its unchanged hypotheses. A further instance already quantified by a theorem does not reopen it. An objection names a failed hypothesis, an invalid bridge, or an inference not preserved; doubt alone leaves a claim OPEN, not FALSE.

Table 7: The claim register.

Claim	Status	Statement	Receipt (Lean names drop the prefix `DIYQC.`)	Axioms
`isomer_lock`	Proved	RotationParity (sigma*epsilon) v: for protons the antisymmetric family rotates evenly and the symmetric family oddly; for deuterons the reverse.	`ExchangePair.isomer_lock`, `ExchangePair.fermion_antisymmetric_even`, `ExchangePair.fermion_symmetric_odd`, `ExchangePair.boson_symmetric_even`, `ExchangePair.boson_antisymmetric_odd`	none
`first_cut_exchange`	Proved	The roles stay distinct, yet no role-blind reading recovers which carrier is first; a labeled reading does. The frame kernel FactorsThrough and the seam kernel Recovers are the same proposition.	`FirstCut.role_blind_reading_cannot_order`, `FirstCut.cut_is_held_not_read`, `FirstCut.labelled_reading_recovers_order`, `FirstCut.factorsThrough_iff_recovers`	none
`lock_is_exact`	Proved	Whatever A sends into the other spin class is zero; in particular the part of a two-site coupling that is the same at both nuclei converts nothing.	`ExchangePair.even_cannot_convert`, `SitePair.sum_cannot_convert`	none
`spin_difference_alone_leaves_sector`	Proved	A v has statistics flip sigma; if it also has statistics sigma it is zero.	`ExchangePair.spin_odd_pos_even_leaves_sector`, `ExchangePair.spin_odd_pos_even_vanishes_in_sector`	none
`key_converts`	Proved	A v keeps statistics sigma, lies in spin class flip epsilon, and has rotation parity sigma*flip epsilon.	`ExchangePair.key_converts`, `SitePair.difference_flips`, `SitePair.difference_odd`	none
`only_the_key_converts`	Proved	s = minus and t = minus: the coupling differs between the nuclei and between the positions.	`ExchangePair.converting_coupling_is_doubly_odd`	none

Claim	Status	Statement	Receipt (Lean names drop the prefix `DIYQC.`)	Axioms
`nonvacuity_hydrogen`	Proved	A uniform field annihilates para; a spin difference without a position difference makes a state of the wrong statistics; a field difference across the molecule makes ortho with odd rotation and leads back to para.	`Model.Spins.uniform_difference_pos_even`, `Model.Spins.gradient_difference_pos_odd`, `Model.Hydrogen.para_is_fermion_pair`, `Model.Hydrogen.para_rotates_evenly`, `Model.Hydrogen.uniform_annihilates_para`, `Model.Hydrogen.spin_difference_without_gradient`, `Model.Hydrogen.symmetricEven_wrong_statistics`, `Model.Hydrogen.gradient_converts_para`, `Model.Hydrogen.orthoOdd_is_symmetric`, `Model.Hydrogen.orthoOdd_rotates_oddly`, `Model.Hydrogen.orthoOdd_is_fermion_pair`, `Model.Hydrogen.gradient_returns_to_para`, `Model.Hydrogen.orthoOdd_nonzero`	propext, Quot.sound
`nonvacuity_deuterium`	Proved	The same three cases for a boson pair: the total-spin-zero ortho state rotates evenly; a field difference makes antisymmetric (para) spin with odd rotation and leads back.	`Model.Deuterium.ortho_is_boson_pair`, `Model.Deuterium.ortho_rotates_evenly`, `Model.Deuterium.uniform_annihilates_ortho`, `Model.Deuterium.spin_difference_without_gradient`, `Model.Deuterium.antisymmetricEven_wrong_statistics`, `Model.Deuterium.gradient_converts_ortho`, `Model.Deuterium.paraOdd_is_antisymmetric`, `Model.Deuterium.paraOdd_rotates_oddly`, `Model.Deuterium.paraOdd_is_boson_pair`, `Model.Deuterium.gradient_returns_to_ortho`, `Model.Deuterium.paraOdd_nonzero`	propext, Quot.sound
`recognition_is_not_conversion`	Proved	A uniform-field readout cannot recognize the isomer; any adequate added coordinate separates the middle ortho state from para; a total-spin reading recognizes it; conversion changes the total spin.	`Recognition.uniform_reading_cannot_recognize`, `Recognition.adequate_coordinate_separates`, `Recognition.total_spin_recognizes`, `Recognition.conversion_changes_total_spin`	propext

Claim	Status	Statement	Receipt (Lean names drop the prefix `DIYQC.`)	Axioms
`sink`	Proved	The rotational energy changes by at least 2B.	`Sink.rotLevel_gap`, `Sink.parity_change_needs_energy`, `Sink.lowest_gap`	propext, Quot.sound
`counting`	Proved	sym d + anti d = d^2; 3:1 for H2, 6:3 for D2; the cold family is a single state exactly for a spin-1/2 fermion pair, and has three or more states for a boson pair with spin.	`Counting.sym_add_anti`, `Counting.hydrogen_families`, `Counting.deuterium_families`, `Counting.cold_family_single_iff`	propext, Quot.sound
`witness`	Proved	Product states and their mixtures have singlet weight <= 1/2; weight above 1/2 excludes them.	`Witness.lagrange`, `Witness.product_singlet_bound`, `Witness.separable_singlet_bound`, `Witness.above_half_is_entangled`	propext, Quot.sound
`two_turn`	Proved	The sign returns after two turns; a direction reading cannot recover it; an adequate reference must separate zero turns from one; under the bridge, proton pairs are fermions and deuteron pairs bosons.	`TwoTurn.returns_after_two_turns`, `TwoTurn.direction_cannot_recover_sign`, `TwoTurn.reference_must_separate`, `TwoTurn.proton_pair_is_fermion`, `TwoTurn.deuteron_pair_is_boson`	propext, Quot.sound
`edge_on`	Proved	Seen in its own plane, the two senses of a rotation give the same swing; the phase is not recovered by any decoder of the swing.	`EdgeOn.fold_symm`, `EdgeOn.edge_on_cannot_recover_phase`	propext, Quot.sound
`closure`	Proved	260 and 365 close at 18980; no positive number of exact fifths closes on octaves; the 2:1 closure behind the selective gate.	`Closure.calendar_round`, `Closure.tzolkin_round`, `Closure.pythagorean_nonclosure`, `Closure.pythagorean_comma`, `Closure.selective_closure`	propext, Quot.sound
`threshold`	Proved	Below threshold (c x < N) the count dies out within x rounds; at or above it, no round lowers it.	`Threshold.step_lt`, `Threshold.below_threshold_dies_out`, `Threshold.above_threshold_no_decrease`	propext, Quot.sound
`B1`	Bridge	Two identical nuclei with spin and rotation form an ExchangePair; total exchange = spin exchange composed with position exchange; the statistics sign comes from spin-statistics.	argued in the paper; SpinStatisticsBridge held as a structure premise	
`B2`	Bridge	Couplings act linearly and decompose into the four parity classes under the two commuting exchanges.	standard linear algebra, stated in the paper	
`B3`	Bridge	A catalyst field at the two nuclei is a SitePair; "differs across the bond" means odd position parity of the difference part.	argued in the paper	

Claim	Status	Statement	Receipt (Lean names drop the prefix DIYQC.)	Axioms
B4	Bridge	Rotational levels B J(J+1); measured levels change the threshold by about 0.1 K.	computations/para_fraction_threshold.py	
B5	Bridge	Equilibrium para fraction from Boltzmann weights with nuclear degeneracies 1:3 (H2), 6:3 (D2).	computations/*.py	
B6	Bridge	Gaussian-integer amplitudes stand for complex amplitudes (homogeneity and density).	argued in the paper	
B7	Bridge	A separable state in finite dimension is a finite mixture of product states.	imported (Carathéodory)	
M1	Bridge	EdgeOn.fold_symm and edge_on_cannot_recover_phase are the index form of Part II inversion_preserves_spectral_read, spectral_collision_has_two_branches and spectral_read_cannot_recover_phase; the step between forms is that cos(2 pi k/n) is injective for 0 <= k <= n/2.	Part II receipts (LambdaPartII.lean, Mathlib); not recompiled here	
H1	Hypothesis	The low-field rate decrease over rare-earth catalysts sits where Zeeman and rotational levels become shared (crossings), where Part II section 8 predicts first-order rather than second-order behavior.	none yet; test: rate against field through the crossing fields	
C1	Computed	Para fraction exceeds 1/2 below 77.84 K (measured levels; 77.94 K rigid rotor). At 77.36 K the equilibrium margin is +0.33 points; pumped LN2 at 63.15-65 K gives +10.5 to +12.3.	computations/para_fraction_threshold.py and results	
C2	Computed	D2 gate-entangleable boundary ortho fraction 6/7 (39.29 K); best negativity (7q-6)/12.	computations/deuterium_spin_order.py and results	
C3	Computed	Two-proton register thresholds: kept order p > 0.5; dephased p > 0.78.	computations/parahydrogen_entanglement_boundary.py and results	
C4	Computed	NV selective CNOT: Omega = A/sqrt(3); rectangular beat Gaussian in 54/54 cases.	computations/nv_cnot_selectivity.py and results	
I1	Imported	Para-H2 (20 K, charcoal) derived two-proton state: purity 0.916, entanglement of formation 0.822; Deutsch and Grover run on such states.	Anwar et al. PRL 93, 040501 (2004); PRA 70, 032324 (2004); arXiv quant-ph/0407091	
I2	Imported	LN2 77 K generator with hydrated iron(III) oxide reaches about 48 % para.	Chapman et al., Anal. Chem. 2021	

Claim	Status	Statement	Receipt (Lean names drop the prefix DIYQC.)	Axioms
I3	Imported	Conversion needs an inhomogeneous field; a homogeneous field should not affect it; rate scales as mu^2 d^-6.	Wigner 1933 via Ilisca, PCCP 2025	
I4	Imported	H2 J=0 to J=1 interval 118.48677 cm^-1.	Beyer et al., PRL 123, 163002 (2019)	
01	Open	Conversion rate magnitudes.	none	
02	Open	Low-field rate decrease over rare-earth catalysts.	Ilisca 2025 reports no explanation	
03	Open	Mechanisms on ferromagnetic and non-magnetic surfaces.	Ilisca 2025	
04	Open	Lean witness for the D2 6/7 boundary; Werner converse direction.	none	
05	Open	Cheapest readout and transfer chemistry for a garage build.	none	

C. Kernel report

```
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'DIYQC.Model.Deuterium.orthoBack' does not depend on any axioms
'DIYQC.Model.Deuterium.ortho_is_boson_pair' depends on axioms: [propext, Quot.sound]
'DIYQC.Model.Deuterium.ortho_is_symmetric' depends on axioms: [propext, Quot.sound]
'DIYQC.Model.Deuterium.ortho_rotates_evenly' depends on axioms: [propext, Quot.sound]
'DIYQC.Model.Deuterium.uniform_annihilates_ortho' depends on axioms: [propext, Quot.sound]
'DIYQC.Model.Deuterium.spin_difference_without_gradient' depends on axioms: [propext, Quot.sound]
'DIYQC.Model.Deuterium.antisymmetricEven_wrong_statistics' depends on axioms: [propext, Quot.sound]
'DIYQC.Model.Deuterium.gradient_converts_ortho' depends on axioms: [propext, Quot.sound]
'DIYQC.Model.Deuterium.paraOdd_is_boson_pair' depends on axioms: [propext, Quot.sound]
'DIYQC.Model.Deuterium.paraOdd_is_antisymmetric' depends on axioms: [propext, Quot.sound]
'DIYQC.Model.Deuterium.paraOdd_rotates_oddly' depends on axioms: [propext, Quot.sound]
'DIYQC.Model.Deuterium.gradient_returns_to_ortho' depends on axioms: [propext, Quot.sound]
'DIYQC.Model.Deuterium.paraOdd_nonzero' depends on axioms: [propext, Quot.sound]
'DIYQC.Recognition.PairState' does not depend on any axioms
'DIYQC.Recognition.uniformReading' does not depend on any axioms
'DIYQC.Recognition.isomer' depends on axioms: [propext]
'DIYQC.Recognition.totalSpinReading' depends on axioms: [propext]
'DIYQC.Recognition.uniform_reading_cannot_recognize' depends on axioms: [propext]
'DIYQC.Recognition.adequate_coordinate_separates' depends on axioms: [propext]
```

```
'DIYQC.Recognition.total_spin_recognizes' depends on axioms: [propext]
'DIYQC.Recognition.conversion_changes_total_spin' depends on axioms: [propext]
'DIYQC.Sink.rotLevel' does not depend on any axioms
'DIYQC.Sink.lowest_gap' does not depend on any axioms
'DIYQC.Sink.rotLevel_gap' depends on axioms: [propext, Quot.sound]
'DIYQC.Sink.parity_change_needs_energy' depends on axioms: [propext, Quot.sound]
'DIYQC.Counting.sym' does not depend on any axioms
'DIYQC.Counting.anti' does not depend on any axioms
'DIYQC.Counting.sym_add_anti' depends on axioms: [propext, Quot.sound]
'DIYQC.Counting.sym_eq_anti_add' depends on axioms: [propext, Quot.sound]
'DIYQC.Counting.hydrogen_families' does not depend on any axioms
'DIYQC.Counting.deuterium_families' does not depend on any axioms
'DIYQC.Counting.coldFamily' does not depend on any axioms
'DIYQC.Counting.anti_ge_three' depends on axioms: [propext, Quot.sound]
'DIYQC.Counting.sym_ge_three' depends on axioms: [propext, Quot.sound]
'DIYQC.Counting.cold_family_single_iff' depends on axioms: [propext, Quot.sound]
'DIYQC.Witness.Amp2' does not depend on any axioms
'DIYQC.Witness.norm2' does not depend on any axioms
'DIYQC.Witness.detRe' does not depend on any axioms
'DIYQC.Witness.detIm' does not depend on any axioms
'DIYQC.Witness.ipRe' does not depend on any axioms
'DIYQC.Witness.ipIm' does not depend on any axioms
'DIYQC.Witness.lagrange' depends on axioms: [propext, Quot.sound]
'DIYQC.Witness.square_nonneg' depends on axioms: [propext, Quot.sound]
'DIYQC.Witness.product_singlet_bound' depends on axioms: [propext, Quot.sound]
'DIYQC.Witness.mixtureSinglet' does not depend on any axioms
'DIYQC.Witness.mixtureNorm' does not depend on any axioms
'DIYQC.Witness.separable_singlet_bound' depends on axioms: [propext, Quot.sound]
'DIYQC.Witness.above_half_is_entangled' depends on axioms: [propext, Quot.sound]
'DIYQC.TwoTurn.spinorSign' does not depend on any axioms
'DIYQC.TwoTurn.directionReading' does not depend on any axioms
'DIYQC.TwoTurn.one_turn_flips' does not depend on any axioms
'DIYQC.TwoTurn.returns_after_two_turns' depends on axioms: [propext, Quot.sound]
'DIYQC.TwoTurn.direction_cannot_recover_sign' does not depend on any axioms
'DIYQC.TwoTurn.reference_must_separate' does not depend on any axioms
'DIYQC.TwoTurn.SpinStatisticsBridge' does not depend on any axioms
'DIYQC.TwoTurn.proton_pair_is_fermion' does not depend on any axioms
'DIYQC.TwoTurn.deuteron_pair_is_boson' does not depend on any axioms
'DIYQC.EdgeOn.fold' does not depend on any axioms
'DIYQC.EdgeOn.fold_symm' depends on axioms: [propext, Quot.sound]
'DIYQC.EdgeOn.edge_on_cannot_recover_phase' depends on axioms: [propext, Quot.sound]
'DIYQC.EdgeOn.quarter_turns_collide' does not depend on any axioms
'DIYQC.EdgeOn.swing_positions_12' does not depend on any axioms
'DIYQC.Closure.tzolkin_round' does not depend on any axioms
'DIYQC.Closure.calendar_round' does not depend on any axioms
'DIYQC.Closure.calendar_round_gcd' does not depend on any axioms
'DIYQC.Closure.calendar_round_counts' does not depend on any axioms
'DIYQC.Closure.selective_closure' depends on axioms: [propext, Quot.sound]
'DIYQC.Closure.three_pow_odd' depends on axioms: [propext, Quot.sound]
'DIYQC.Closure.two_pow_even' depends on axioms: [propext, Quot.sound]
'DIYQC.Closure.pythagorean_nonclosure' depends on axioms: [propext, Quot.sound]
'DIYQC.Closure.pythagorean_comma' does not depend on any axioms
'DIYQC.Threshold.step' does not depend on any axioms
'DIYQC.Threshold.iter' does not depend on any axioms
'DIYQC.Threshold.step_zero' depends on axioms: [propext]
'DIYQC.Threshold.step_lt' depends on axioms: [propext]
'DIYQC.Threshold.iter_bound' depends on axioms: [propext, Quot.sound]
'DIYQC.Threshold.below_threshold_dies_out' depends on axioms: [propext, Quot.sound]
'DIYQC.Threshold.above_threshold_no_decrease' depends on axioms: [propext]
```

D. The record of its own making

Search before building. The corpus repository [8] was read for Lean kernels that were already certified before anything was written. Ninety-nine Lean files were collected, 61 of them distinct. Three modules that the new statements could stand on were reused unchanged instead of being proved again. Every collision result in the kernel is an application of those modules, and the frame and seam formulations were shown to be one proposition.

The toolchain. The release archive of Lean 4.35.0 could not be downloaded from the build

environment, so Lean was compiled from the official source tree at the commit recorded in the Axiom as Pharmakon R3 receipt [5]. While the release compiler built, the bootstrap compiler of the same tree was used for trial checks only. Every result reported here comes from the release compiler, through the audit script of Appendix A.

Errors found and fixed (recorded once).

1. *A physics error in the first draft of the model.* It used one rotational factor for both the uniform part and the difference part of a two-site coupling. That made the uniform part odd under position exchange, which no coupling of two identical particles can be. The model was rebuilt with two site pairs: a uniform field with no rotational factor, and a field difference with a factor that changes the parity of J. This change also exposed the question that Theorems 4(ii) and 5 answer: why the difference must differ in position as well.
2. *Compile-stage corrections, with no statement changed.* One lemma's arguments are implicit in the library (`Int.neg_add`). The linter flagged three unused simplification lemmas, and they were removed. A core lemma assumed present (`Int.mul_self_nonneg`) is absent, so it was replaced by a proved one (`Witness.square_nonneg`). In the one-paste bundle, universe names repeat across files, so each file is wrapped in its own section.
3. *Computations.* The earlier scripts carry their own error records. In one, a flat region stalled the D_2 optimizer, and a two-stage search fixed it. In another, an exploratory check had a rotational temperature halved.
4. *The threshold model.* The rigid rotor gives 77.94 K; the measured levels give 77.84 K. Both are reported (C1).
5. *Coverage.* The two-deuteron model was added after the two-proton model so that both statistics are witnessed.

The figure. The labels of Figure 1 were moved three times to remove collisions; the data did not change.

This document. `build_paper.py` fills every number, hash and kernel report in the text from the receipts and the computation outputs, then runs three XeLaTeX passes. The build fails on any line more than 1 pt too wide, any missing glyph and any undefined reference. The LaTeX source, the figure script, the build script and the kernel report are attached to this PDF. A final proofreading pass, at the author's invitation, corrected spelling and capitalization to the corpus's US usage. The claim register in Appendix B is rendered with the same spelling; the hosted supplement files keep their original text, so the hashes in Appendix A stand.

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